

Topic: Car Rental / Vehicle Fleet Management Database

Description: The Car Rental and Vehicle Fleet Management Database is a centralised relational database system designed to streamline day-to-day operations for an automobile rental business. Managing a modern vehicle fleet requires tracking multiple moving parts, including asset availability, customer verification, rental agreements, and maintenance lifecycles. This project implements a structured database architecture that ensures data consistency, eliminates redundancy, and optimises operational efficiency.

At its core, the schema organises data into four critical business segments: inventory management (Cars), customer records (Customers), transactional operations (Bookings), and asset protection (Maintenance_Logs). It models real-world logical constraints, such as ensuring a vehicle cannot be double-booked and tracking driver license details for legal compliance.

By progressing through the full database lifecycle, this project demonstrates end-to-end relational database management capabilities. It moves from initial requirement analysis and entity-relationship mapping to schema implementation and structural optimisation. Advanced database features like stored procedures automate recurring tasks like booking cost computations, while triggers automatically update car availability flags to secure data integrity. Ultimately, the system serves as a scalable foundation for business analytics, tracking fleet utilisation rates and maintenance costs to maximise profitability.

Entities required:

Entities1(Cars)

Entities2(Customers)

Entities3(Bookings)

Entities4(Maintenance_Logs)

Entities	Attributes
Cars	car_id, (Unique Identifier / Primary Key) make (e.g., Toyota, Honda, Ford) model (e.g., Camry, Civic, Mustang) year, license_plate, (Must be unique) daily_rate, (The rental price per day) status (Available, Rented, In Maintenance)
Customers	customer_id (Unique Identifier / Primary Key) first_name last_name email (Must be unique) phone_number driver_license_number (Crucial for a

	rental system)
Bookings	booking_id (Unique Identifier / Primary Key) customer_id (Links back to the Customers table) car_id (Links back to the Cars table) start_date end_date total_cost
Maintenance_Logs	log_id (Unique Identifier / Primary Key) car_id (Links back to the specific Car that was serviced) service_date description (e.g., "Oil change and brake replacement") cost

